MTH313M PRACTICE PROBLEMS WEEK 2

Question 1. Let $\{N_t\}_{t\geq 0}$ be a Poisson Process with rate λ. For any $s < t$ in $[0, \infty)$ and non-negative integers $k \leq n$, show that

$$\mathbb{P}(N_s = k \mid N_t = n) = \binom{n}{k} \left(\frac{s}{t}\right)^k \left[1 - \frac{s}{t}\right]^{n-k}.$$

Question 2. Let $S_1, S_2, \cdots$ be the arrival times of the events observed in a Poisson Process $\{N_t\}_{t\geq 0}$ with rate λ. Find the conditional joint distribution function $F_{S_1,S_2|N_t=2}$. [Hint: Use the conditional joint p.d.f. discussed in class]

Question 3. Let $S_1, S_2, \cdots$ be the arrival times of the events observed in a Poisson Process $\{N_t\}_{t\geq 0}$ with rate λ. Find the conditional joint p.d.f. $f_{S_1,S_2,\cdots,S_k|N_t=k}$ and hence find the conditional joint distribution function $F_{S_1,S_2,\cdots,S_k|N_t=k}$. [Hint: complete the argument discussed in in class]

Question 4. Let $S_1, S_2, \cdots$ be the arrival times of the events observed in a Poisson Process $\{N_t\}_{t\geq 0}$ with rate λ. The following approach can be used to find the conditional joint distribution function $F_{S_1,S_2,\cdots,S_k|N_t=k}$ directly.

$$F_{S_1,S_2,\cdots,S_k|N_t=k}(s_1, s_2, \cdots, s_k) = \frac{\mathbb{P}(S_1 \leq s_1, S_2 \leq s_2, \cdots, S_k \leq s_k, N_t = k)}{\mathbb{P}(N_t = k)}$$

Here, the numerator can be simplified as follows:

$$\begin{aligned}
&\mathbb{P}(S_1 \leq s_1, S_2 \leq s_2, \cdots, S_k \leq s_k, N_t = k) \\
=&\mathbb{P}(T_1 \leq s_1, T_1 + T_2 \leq s_2, \cdots, T_1 + T_2 + \cdots + T_k \leq s_k, T_1 + T_2 + \cdots + T_k + T_{k+1} > t) \\
=&\int_{u_1=0}^{s_1} \int_{u_2=0}^{s_2-u_1} \cdots \int_{u_k=0}^{s_k-(u_1+\cdots+u_{k-1})} \int_{u_{k+1}=t-(u_1+\cdots+u_k)}^{\infty} \lambda^{k+1} \exp(-\lambda(u_1 + \cdots + u_k + u_{k+1}))\, du_{k+1}\, du_k \cdots du_1
\end{aligned}$$

Complete the argument.

Question 5. Suppose arrivals in a system happen with $\{N_t\}_{t\geq 0}$ following a PP with rate $\lambda > 0$. A device registers the arrivals with success rate $p \in (0, 1)$, independently. Let the process $\{X_t\}_{t\geq 0}$ denote the registered arrivals. Show that $\{X_t\}_{t\geq 0}$ is a PP with rate λp.

Question 6. Assume that letters arrive at a post office following a PP with a rate 3 letters per hour. What is the probability that no letters shall arrive during the morning hours $8:00$ am to $12:00$ noon?

Question 7. Complete the exercises left out in class.